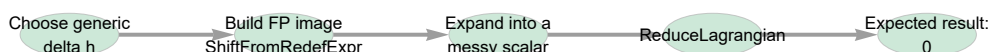


Graviton Basis Toolkit Validation Report

This notebook is the notebook and PDF rendering of `graviton_basis_toolkit_validation_report.md`. It documents the validation strategy and records the current test outcome.



Validation Strategy

The reducer is tested on expressions that are guaranteed to be redundant: images of the Fierz-Pauli kinetic term under nonlinear field redefinitions. In EFT language these are EOM-proportional terms up to a total derivative.

- Choose a generic rank-2 redefinition from the computed `RedefDeltaBasis`.
- Map it into the target scalar sector with `ShiftFromRedefExpr`.
- Expand the result into a large scalar expression.
- Reduce the expression with `ReduceLagrangian`.
- Require the reduced representative to vanish exactly.

Current Results

The table below comes from `graviton_basis_toolkit_validation_output.wl`.

Sector	IBPCount	RedefRank	PhysicalCount	DeltaBasisLength	LagTermCount	Reduced to 0?
{3, 2}	14	4	10	4	18	True
{3, 4}	64	33	31	33	240	True

Interpretation

The {3,2} test is a compact sanity check. The {3,4} test is the stronger one because it uses all 33 rank-2 redefinition basis elements in that sector and expands to 240 scalar terms before reduction.

Re-run Command

```
& 'C:\Program Files\Wolfram Research\Wolfram\14.2\WolframKernel.exe' -script graviton_ba
```

After rerunning, inspect `graviton_basis_toolkit_validation_output.wl` for the updated association.

Files Produced by the Validation

- `graviton_basis_toolkit_validation.wls`: validation script
- `graviton_basis_toolkit_validation_output.wl`: machine-readable output
- `graviton_basis_validation_cache/`: persistent cache used by the validation run